

X1
DigitalCarrierJack.SchDoc

Encoder1
Encoder1.SchDoc

Encoder2
Encoder2.SchDoc

Encoder3
Encoder3.SchDoc

Connector
Connectors.SchDoc

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

Encoder_Signals_1
Encoder_Signals_2
Encoder_Signals_3

Encoder_Signals

Encoder_Signals

Encoder_Signals

GND
GND



Revision

Serial
Serial

UltraZohm_DigitalVoltage
Project

Designer

Date

Title		TopSheet.SchDoc		Technische Hochschule Nürnberg Georg Simon Ohm Fürther Straße 250 90429 Nürnberg GERMANY			
Size:	A4	Revision:	1v00				
Date:	02.03.2022	Time:	15:47:01	Sheet 1	of 6		TECHNISCHE HOCHSCHULE NÜRNBERG GEORG SIMON OHM
Project:	UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB						
				Author: Michaela Hlatky			

A

B

C

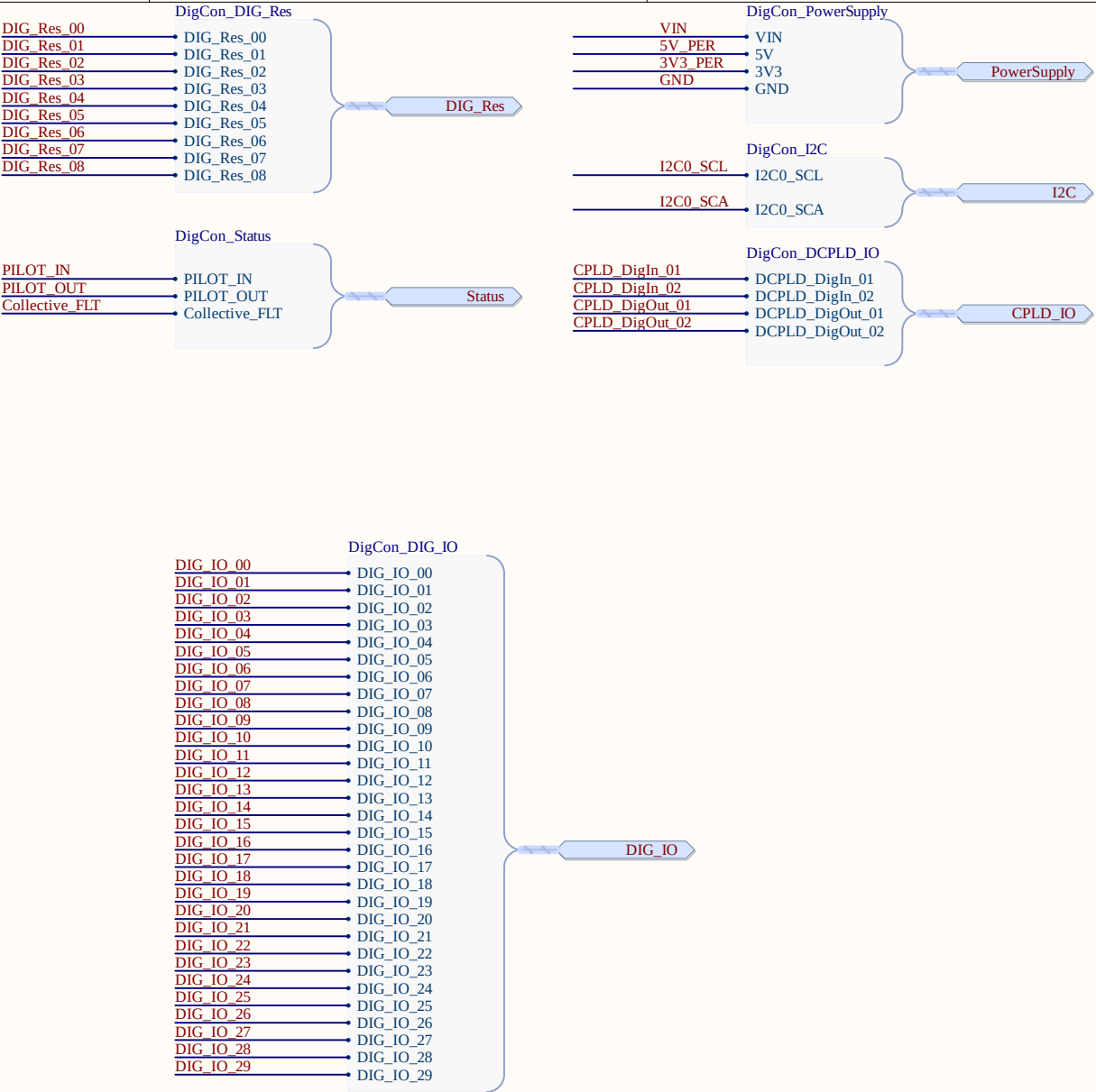
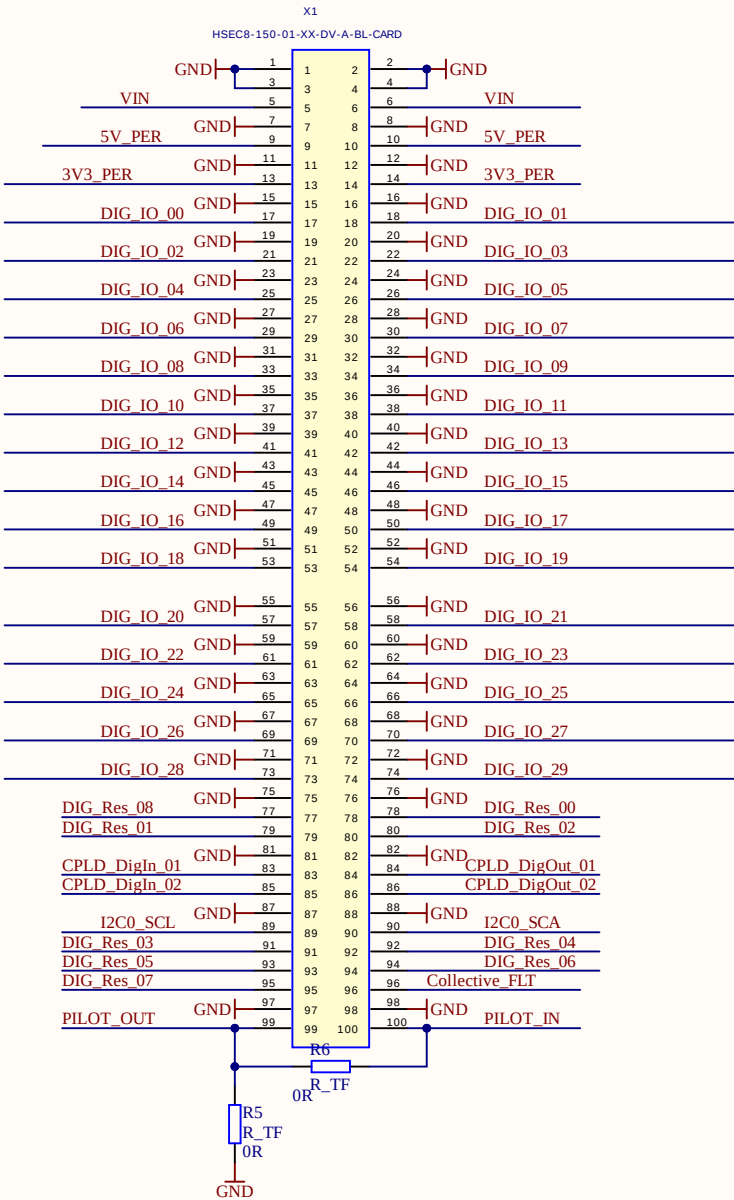
D

A

B

C

D



A

B

C

D

A

B

C

D

DigCon_DIG_IO

DIG_IO_00	DIG_IO_00
DIG_IO_01	DIG_IO_01
DIG_IO_02	DIG_IO_02
DIG_IO_03	DIG_IO_03
DIG_IO_04	DIG_IO_04
DIG_IO_05	DIG_IO_05
DIG_IO_06	DIG_IO_06
DIG_IO_07	DIG_IO_07
DIG_IO_08	DIG_IO_08
DIG_IO_09	DIG_IO_09
DIG_IO_10	DIG_IO_10
DIG_IO_11	DIG_IO_11
DIG_IO_12	DIG_IO_12
DIG_IO_13	DIG_IO_13
DIG_IO_14	DIG_IO_14
DIG_IO_15	DIG_IO_15
DIG_IO_16	DIG_IO_16
DIG_IO_17	DIG_IO_17
DIG_IO_18	DIG_IO_18
DIG_IO_19	DIG_IO_19
DIG_IO_20	DIG_IO_20
DIG_IO_21	DIG_IO_21
DIG_IO_22	DIG_IO_22
DIG_IO_23	DIG_IO_23
DIG_IO_24	DIG_IO_24
DIG_IO_25	DIG_IO_25
DIG_IO_26	DIG_IO_26
DIG_IO_27	DIG_IO_27
DIG_IO_28	DIG_IO_28
DIG_IO_29	DIG_IO_29

DIG_IO

Encoder_Signals_CH3

A_encoder	A_encoder3
B_encoder	B_encoder3
Z_encoder	Z_encoder3

Encoder_Signals_3

Encoder_Signals_CH2

A_encoder	A_encoder2
B_encoder	B_encoder2
Z_encoder	Z_encoder2

Encoder_Signals_2

Encoder_Signals_CH1

A_encoder	A_encoder1
B_encoder	B_encoder1
Z_encoder	Z_encoder1

Encoder_Signals_1

DIG_IO_00	01	02	02 DIG_IO_02
DIG_IO_01	04	05	05 DIG_IO_04
DIG_IO_03	07	08	08 DIG_IO_06
DIG_IO_05	10	11	11 DIG_IO_08
DIG_IO_07	13	14	14 DIG_IO_10
DIG_IO_09	16	17	17 DIG_IO_12
DIG_IO_11	19	20	20 DIG_IO_14
DIG_IO_13	22	23	23 DIG_IO_16
DIG_IO_15	25	26	26 DIG_IO_18
DIG_IO_17	28	29	29 DIG_IO_19

GND

X2

03

06

09

12

15

18

21

24

27

30

Pin Header

X4

VIN_fused	01	02	02 VIN_fused
GND	03	04	GND
5V_PER_fused	05	06	06 5V_PER_fused
GND	07	08	GND
3V3_PER_fused	09	10	10 3V3_PER_fused
GND	11	12	GND

Pin Header

DigCon_DIG_Res

DIG_Res_00	DIG_Res_00
DIG_Res_01	DIG_Res_01
DIG_Res_02	DIG_Res_02
DIG_Res_03	DIG_Res_03
DIG_Res_04	DIG_Res_04
DIG_Res_05	DIG_Res_05
DIG_Res_06	DIG_Res_06
DIG_Res_07	DIG_Res_07
DIG_Res_08	DIG_Res_08

DIG_Res

DigCon_DCPLD_IO

DCPLD_DigIn_01	CPLD_DigIn_01
DCPLD_DigIn_02	CPLD_DigIn_02
DCPLD_DigOut_01	CPLD_DigOut_01
DCPLD_DigOut_02	CPLD_DigOut_02

CPLD_IO

DigCon_Status

PILOT_IN	PILOT_IN
PILOT_OUT	PILOT_OUT
Collective_FLT	Collective_FLT

Status

DIG_IO_21	01	02	02 DIG_IO_20
DIG_IO_23	04	05	05 DIG_IO_22
DIG_IO_25	07	08	08 DIG_IO_24
DIG_IO_27	10	11	11 DIG_IO_26
DIG_IO_29	13	14	14 DIG_IO_28
DIG_Res_07	16	17	17 DIG_Res_08
CPLD_DigIn_01	19	20	20 CPLD_DigOut_01
CPLD_DigIn_02	22	23	23 CPLD_DigOut_02
I2C0_SCL	25	26	26 I2C0_SCA
PILOT_OUT	28	29	29 PILOT_IN

GND

X3

03

06

09

12

15

18

21

24

27

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Pin Header

F1

VIN	VIN_fused
Littelfuse_PTC_1812	
F2	
5V_PER	5V_PER_fused
Littelfuse_PTC_1812	
F3	
3V3_PER	3V3_PER_fused
Littelfuse_PTC_1812	

DigCon_I2C

I2C0_SCL	I2C0_SCL
I2C0_SCA	I2C0_SCA

I2C

DigCon_PowerSupply

VIN	VIN
5V	5V_PER
3V3	3V3_PER
GND	GND

PowerSupply

Title Connectors.SchDoc

Size: A4

Revision: 1v00

Date: 02.03.2022 Time: 15:47:01

Sheet 3 of 6

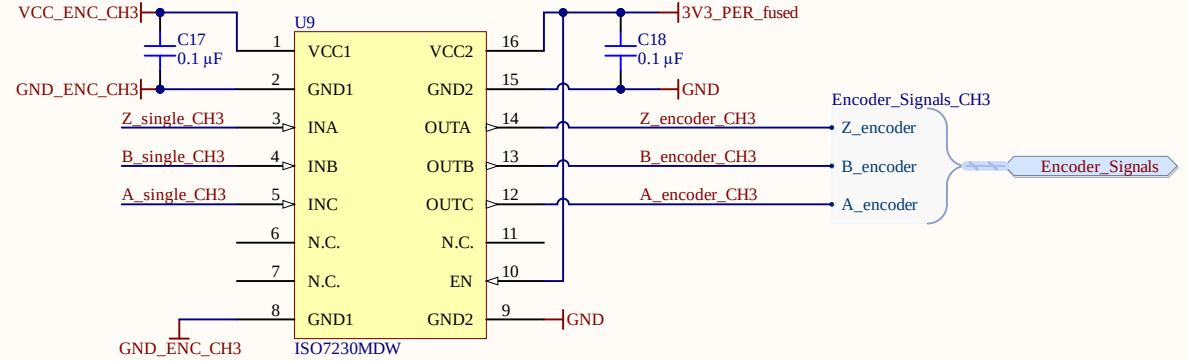
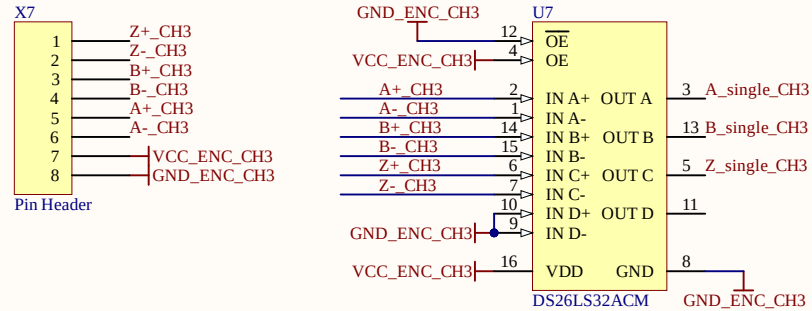
Author: Michaela Hlatky

Project: UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB

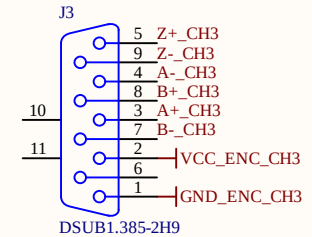
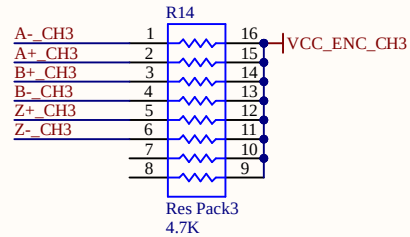
Technische Hochschule Nürnberg Georg Simon Ohm
Fürther Straße 250
90429 Nürnberg
GERMANY

TECHNISCHE HOCHSCHULE NÜRNBERG
GEORG SIMON OHM

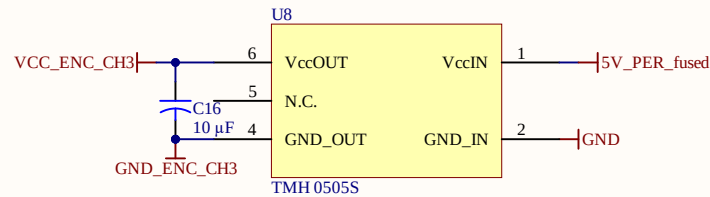
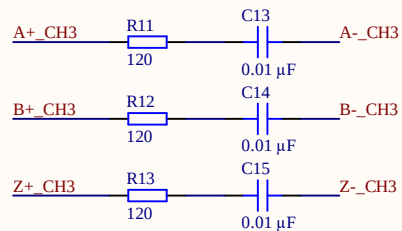
A



B

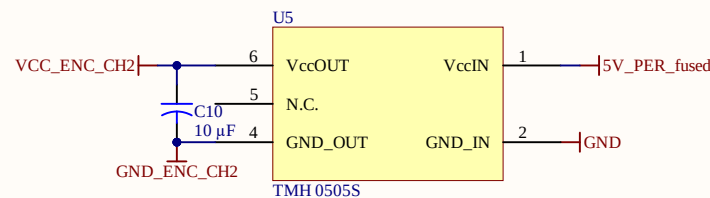
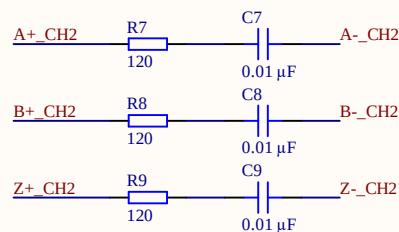
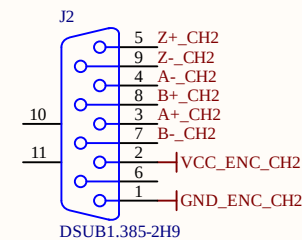
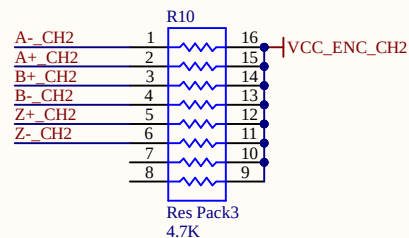
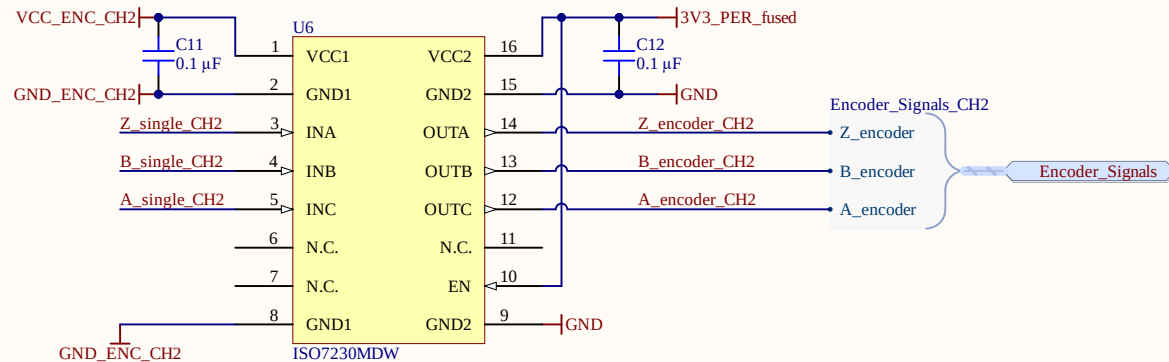
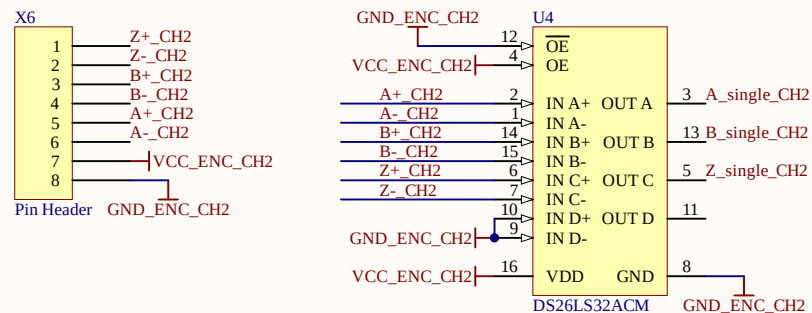


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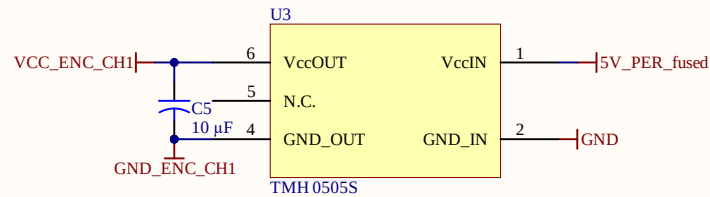
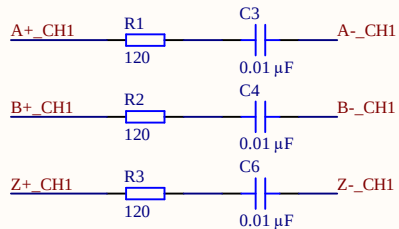
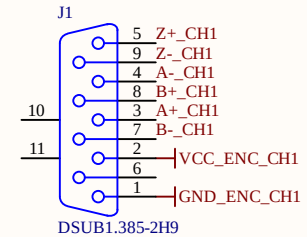
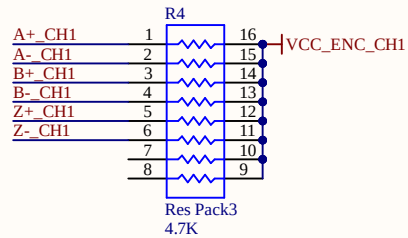
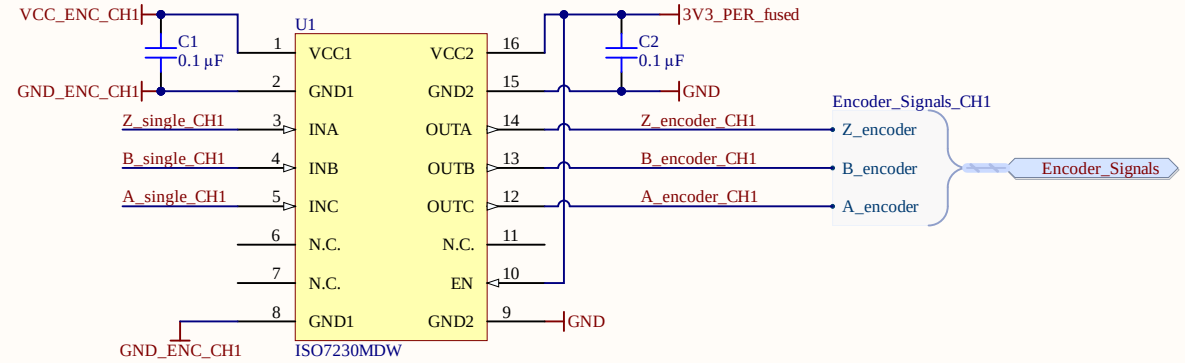
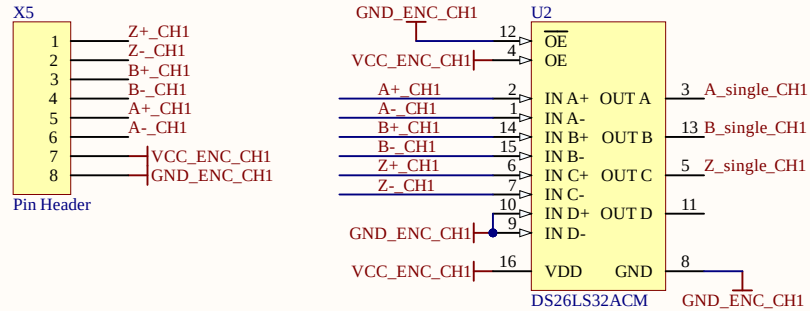


Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	

D



Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	



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U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	